Enhancing Telemedicine Security: A Novel Certificateless Aggregate Signature Scheme with Revocation Management

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***Abstract*—With the integration of a telemedicine system network, the efficiency and quality of medical services have seen significant improvement. However, the utilization of wireless communication for data transmission and storage in telemedicine networks exposes users' privacy to various threats. The dissemination of disinformation and attacks within these networks can result in severe damage and compromise the overall security of the healthcare system. To address these challenges, this paper proposes a novel certificateless aggregate signature scheme with revocation management. This scheme achieves rapid and efficient resource aggregation, enhances telemedicine security, and reduces the overhead of data transmission calculations. Incorporating a cuckoo filter, trust evaluation, and a cloud server enables cross-domain authentication and designated revocation. Comprehensive analysis confirms the protocol's high communication security and computational efficiency.**

***Index Terms*—Telemedicine; wireless medical sensor network; certificateless aggregate signature; cuckoo filter; trust assessment mechanism**

I. Introduction

Telemedicine [1], also known as telehealth or e-health, is a form of health care that uses advanced information and communication technologies to deliver medical services and information to patients who are far away from a medical facility. Telemedicine can help allocate medical resources more flexibly, reduce the burden on traditional medical institutions, and improve the efficiency of medical services. Patients are no longer restricted by geographical location and can obtain more convenient medical services through telemedicine.

Unfortunately, medical communication sensor nodes [2] play a crucial role in telemedicine by monitoring patients' real-time physiological data. Ensuring the accuracy of shared medical data before diagnosis is essential. While aggregate signature schemes [3] in smart healthcare [4] facilitate secure data sharing, the inefficiencies introduced by bilinear operations [5] pose challenges for large-scale deployment. Additionally, the digital storage of personal health information raises privacy concerns. Consequently, contemporary telemedicine systems prioritize patient-controlled medical information to enhance security and privacy-focused services.

Therefore, this paper proposes a certificateless aggregate signature authentication scheme with specified revocation. Leveraging certificateless aggregate signature technology, the scheme simplifies patient identity verification, thereby reducing system complexity [6]. Integrated with cross-domain authentication of edge servers, this approach ensures the security and integrity of patient medical data in telemedicine systems. The user revocation function empowers patients to actively manage and adjust information-sharing permissions, meeting evolving privacy needs. This capability allows patients to revoke access rights at any time, enhancing their awareness of personal privacy and reinforcing trust in the telemedicine system.

Section II of this paper provides an overview of the literature review and related work. Section III introduces the scheme proposed in this paper in detail. In Section IV, the security analysis and performance analysis of the proposed method are presented. Section V provides conclusions.

II. Related work

Boneh et al. [3] proposed a short signature scheme based on the bilinear pairing. Subsequently,Anna et al. [7] constructed a sequential aggregate signature scheme based on the trapdoor permutations. Castro et al. [8] designed a novel certificateless signature scheme that is suitable for aggregation. Xiong et al. [9] presented an efficient certificateless aggregate signature scheme with the constant pairing computations.

Gayathri et al. [10] and Kumar et al. [11] suggested a certificateless aggregated signature. Mei et al. [12] offered a certificateless aggregated signature scheme with conditional privacy protection in vehicular networks. Zhong et al. [13] made a new privacy-preserving authentication scheme that achieves full aggregation in VANETs. Deng et al. [14] put forward a lightweight certificateless aggregation signature scheme with provably security in the standard model.Shen et al. [15] presented a privacy-preserving and conspiracy-resistant data aggregation scheme designed specifically for fog computing environments.

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Ren et al. [16] developed an identity-based proxy aggregation signature scheme that focuses on efficient signature aggregation using identity-based cryptography. Khan et al. [17] introduced a fog-based privacy-preserving secure data aggregation scheme that incorporates fault tolerance measures. Gowri et al. [18] put forward an aggregate signature scheme based on non-authentication, addressing certificate management and key escrow issues. Ramadanet al. [22] proposed a secure WBAN-19 telemedicine system, leveraging a novel identity-based aggregate signcryption scheme with equal testing capabilities to address the prevalence of infectious diseases.

However, it's crucial to note that none of the mentioned scenarios addresses the issue of revoking user privacy information. Despite focusing on privacy, security, and data aggregation in different contexts, these schemes don't explicitly tackle the challenge of revoking and managing malicious users within the system.

Therefore, Zhou et al. [19] proposed an enhanced security revocable certificateless aggregate signature (CLAS) scheme. Yue et al. [20] developed an aggregate anonymous scheme for multi-service revocation in cooperative intelligent transportation systems. Chen et al. [21] suggested a parameter-free revocable certificateless signature scheme based on blockchain, generating time keys with cloud computing to revoke users with low computational cost and ensuring anonymity.

Yet, none of the above solutions provides a mechanism for users to actively participate and adjust their privacy information sharing permissions at any time.

The system is divided into six basic modules, namely Key Generation Center (KGC), Data Owner (DO), Medical Center (HA), Medical Communication Sensing Node (SN), Data Center (DC) and Medical Server (ID) as shown in Fig.1.

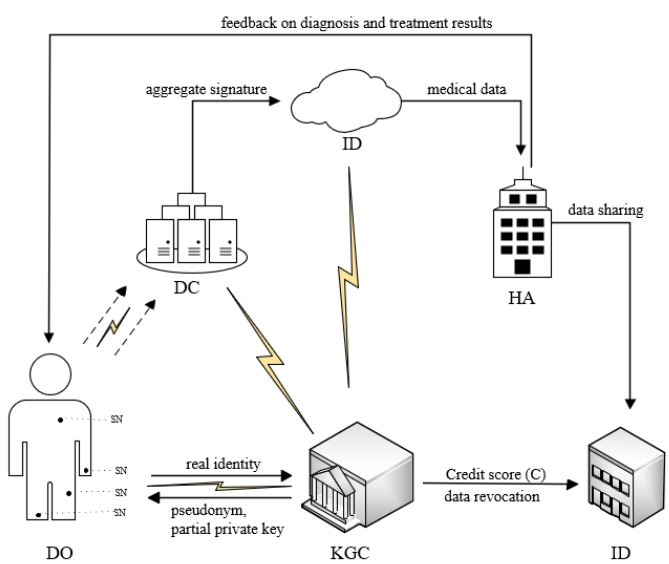

Fig. 1. System model

- **KGC**: KGC is responsible for generating partial private keys for each vehicle and storing their corresponding pseudo identity.
- **DO**: As a medical patient, DO is equipped with n medical sensor nodes and registers with KGC before entering the medical system. KGC can track the true identity of DO.
- **HA**: HA receives patient registration forms from KGC, establishes a credit list, and verifies the ID's aggregated signature. It then analyzes sensing data and provides relevant feedback.
- **SN**: Each DO has n sensor nodes, which are responsible for collecting medical data and transmitting the data to the data center through IoT smart devices.
- **DC**: DC is responsible for collecting signatures from various medical sensing nodes, generating an aggregate signature and sending it to the medical server.
- **ID**: ID manages medical cloud services, records medical data from IoT devices, and creates cuckoo filters. It also serves as a verifier, validating aggregated signatures. In case of malicious behavior, ID collaborates with HA to track and revoke the DO's true identity.

III. The Proposed Scheme

A. System initialization

Before the system runs, HA and KGC initialize the system.

1) HA and KGC chooses the additive cyclic group G with prime order $q(q > 2^{\lambda})$, where λ is a security parameter and P is a generator of G.

2) HA chooses a number $s \in Z_q^*$ randomly as its main private key, computes $T_{pub} = sP$ as its main public key. KGC chooses a random number $k \in Z_q^*$ as its main private key, it computes $K_{pub} = kP$ as its main public key.

3) KGC choose three hash functions: $H_1 : G^2 \times \{0,1\}^* \rightarrow Z_q^*$, $H_2 : \{0,1\}^{*2} \times G^2 \rightarrow Z_q^*$, $H_3 : \{0,1\}^{*4} \times G \rightarrow Z_q^*$. KGC publishes the public parameters $params = \{q, G, P, T_{pub}, K_{pub}, H_1, H_2, H_3\}$. Register DC and users in the initial stage and obtain system parameters in a secure way. HA creates an empty database to store the information of each vehicle to facilitate subsequent uploading of malicious vehicle information to the cloud. ID creates an empty cuckoo filter.

B. User registration

By performing the following operations, DO registers with KGC and generates a public-private key pair.

1) HA randomly chooses $d_i \in Z_q^*$, computes $NID_{i,1} = d_iP, D_i = d_iT_{pub}$ and $NID_{i,2} = RID_i \oplus H_1(r \cdot NID_{i,1}, D_i, T_i)$, sends the pseudo identity $PID_i = (NID_{i,1}, NID_{i,2}, T_i)$ to KGC.

2) KGC chooses a random number $n_i \in Z_q^*$ and calculates $N_i = n_iP$,$h_{2i} = H_2(FID_i, K_{pub}, N_i, T_i)$. Then, KGC can calculate the partial private key $ppk_i = n_i + kh_{2i}$, and sends (ppk_i, PID_i, N_i) to DO via a secure channel.

3) DO selects a random number $s_i \in Z_q^*$ as the secret value and computes $S_i = s_iP$, $H_2 : \{0,1\}^* \times G \times G \times \{0,1\}^* \rightarrow Z_q^*$, $U_i = h_{2i}S_i + N_i$. Then, the private key of DO is

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$SK_i = ppk_i + h_{2i}S_i$ and its corrsponding public key is $PK_i = (U_i, N_i)$.

C. Signature generation and verification

By executing the following algorithm, DO can generate a signature for the message m_i, and DC can verify the signature.

1) DO selects a pair of (ppk_i, PID_i) and computes a hash value $h_{3i} = H_3(m_i, PID_i, PK_i, \nabla, t_i)$, DO chooses a random number $r_i \in Z_q^*$, calculates $R_i = r_iP$, $Y_i = r_i + h_{3i}SK_i$. DO generates the signature $\delta_i = (R_i, Y_i)$ based on the above calculations. DO broadcasts the message tuple $mt = \{PID_i, m_i, \delta_i, t_i\}$ to nearby DC.

2) The DC lookup cuckoo filter ensures the patient has not withdrawn. If not revoked, DC calculates two hash values: $h_{2i} = H_2(PID_i, K_{pub}, N_i, T_i)$ and $h_{3i} = H_3(m_i, PID_i, PK_i, \nabla, t_i)$, and then verifies the equation

$$Y_iP \stackrel{?}{=} R_i + h_{3i}(U_i + h_{2i}K_{pub}) \tag{1}$$

If this equation is true, DC accepts the signature. Otherwise, it aborts.

D. Aggregate signature generation

By performing the following operations, the data center DC can generate an aggregate signature.

1) Assuming there are $\{\delta_1, \delta_2, \cdots, \delta_l\}$ signatures, generated by $\{PID_1, PID_2, \cdots, PID_l\}$ of l SNs for $\{m_1, m_2, \cdots, m_l\}$ messages, that need to be aggregated. DC acts as the aggregator. DC randomly generates a vector $W = \{W_1, W_2, \cdots, W_l\}$, where each $W_i (i = 1, 2, \cdots, l)$ is a random number in the range $[1, 2^{\varpi}]$.

2) DC calculates $Y = \sum_{i=1}^{l} W_iY_i$, $R = \sum_{i=1}^{l} W_iR_i$. Aggregated signature is denoted as $\delta = (R, Y)$ and sent $\delta = (R, Y)$ to ID by DC.

E. Aggregate signature verification

The medical server ID first passes the cuckoo filter to check that the fake identity has not been revoked. If it has not been revoked, the legitimacy of the aggregate signature $\delta = (R, Y)$ can be verified.

1) ID computes the two required hash values: $h_{2i} = H_2(PID_i, K_{pub}, N_i, T_i)$ and $h_{3i} = H_3(m_i, PID_i, PK_i, \nabla, t_i)$, and verifies whether the equation

$$YP \stackrel{?}{=} R + \sum_{i=1}^{l} W_ih_{3i}(U_i + h_{2i}K_{pub}) \tag{2}$$

holds. If the equation holds, ID accepts all signatures δ.

2) If the equation does not hold,ID uses bitwise matching to identify and track out invalid signatures. HA calculates and traces back the real identity $RID_i = NID_{i,2} \oplus H_1(r \cdot NID_{i,1}, D_i, T_i)$ corresponding to the pseudo identity that issued the invalid signature, and inserts it into the cuckoo filter to revoke it.Reduce the reputation score of malicious patients according to the formula $C = C \times b(0 < b < 1)$, and record the updated reputation score in the cuckoo filter. When the patient's reputation score C is less than a certain threshold Y, the patient is removed from the list and added to the blacklist, and the blacklist is published in the medical server ID.

F. Data sharing

The medical server ID records the hash digest value, aggregate signature, and timestamp of patient medical data, ensuring timely updates. Published alongside the cuckoo filter, this information undergoes initial verification in the cuckoo filter upon receipt. Successfully verified data is permanently stored on ID, facilitating data sharing. Access rights for other HAs are managed through the access control protocol. In accessing patient DO's medical records, HA can employ the cuckoo filter to identify and retrieve downloaded medical data, streamlining the data-sharing process.

G. Data revocation

When a patient decides to revoke an institution's access, they initiate a request specifying the institution and use their private key to generate a revocation signature containing relevant details and a timestamp. Cuckoo filters efficiently validate these requests, speeding up checks for institutions on the revocation list. Before sending the patient's revocation signature, the institution's trust score is determined through an evaluation of historical behavior, patient feedback, qualifications, and other factors, ensuring legitimacy. During revocation, the patient's medical information is securely stored on a cloud server for data security and reliability. Upon completion, the system notifies the patient of the revocation status, specifying which institutions had access revoked and when. Following revocation, the institution loses access to new patient medical information.

IV. Performance analysis

A. Security Analysis

Theorem 1: In the ROM, if we consider a Type-1 adversary $\mathcal{A}_\infty$ who, after making a maximum of q_h queries to the random oracles h_iQuery, q_c queries to Vehicle create, and q_s queries to Signature, can forge a valid signature in game 1 with a nonnegligible probability of success $Pr\left[Succ(\mathcal{A}_\infty)\right]$, then it implies the existence of a polynomial time challenger $\mathcal{C}$ who can solve this paper with a significant advantage ε.

Proof: Suppose that there exists a type-1 Probabilistic Polynomial-Time (PPT) adversary $\mathcal{A}_\infty$ who can forge a signature that satisfies the proposed scheme with a non-negligible probability ε. This paper will construct a simulator $\mathcal{B}$ that can simulate the challenges of $\mathcal{C}$. $\mathcal{B}$ randomly selects a challenge anonymous identity FID_i^*. Assuming $\mathcal{A}_\infty$ in the ROM forges the message tuple $mt = \{PID_i, m_i, \delta_i, t_i\}$ for PID_i^*, this problem give an instance $(P, P_0 = aP)$ of the ECDLP to compute a. $\mathcal{A}_\infty$ performs q_H queries to H_iQuery, q_C queries to create vehicle (PID_i), and q_S queries to sign (PID_i, m_i). $\mathcal{A}_\infty$ makes the following queries:

Initialization: $\mathcal{B}$ generate the system parameter $params = \{q, G, P, T_{pub}, K_{pub}, H_1, H_2, H_3\}$, it constructs five lists $L_{H_1}, L_{H_2}, L_{H_3}, L_u, L_{PK}$, and initialize them to empty.

Vehicle creation query: $\mathcal{B}$ manages a list L_u . When $\mathcal{A}_\infty$ initiates an query on PID_i to $\mathcal{B}$, $\mathcal{B}$ make the following query: If the PID_i is in L_u, $\mathcal{B}$ returns $(PID_i, N_i, PK_i, ppk_i, SK_i, h_{2i})$ to $\mathcal{A}_\infty$. Otherwise, $\mathcal{B}$

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chooses $x_i, y_i, z_i \in Z_q^*$, and calculates $N_i = x_i K_{pub} + x_i y_i P$, $PK_i = x_i z_i P + x_i y_i + x_i K_{pub}$, $ppk_i = x_i z_i$, $SK_i = x_i y_i$ and $h_{2i} = H_2(PID_i, K_{pub}, N_i, T_i) \leftarrow -x_i$. Finally, $\mathcal{B}$ returns $(PID_i, N_i, PK_i, ppk_i, SK_i, h_{2i})$ to $\mathcal{A}_\infty$.

H_1**Query**: $\mathcal{B}$ maintains a list L_{H_1} . When $\mathcal{A}_\infty$ requests an H_1 query: $\mathcal{B}$ checks if it is present in L_{H_1}. If not, $\mathcal{B}$ generates h_{1i} value at random, sets $h_{1i} = H_1(s \cdot NID_{i,1}, D_i, T_i)$, returns h_{1i} to $\mathcal{A}_\infty$.

H_2**Query**: $\mathcal{B}$ maintains a list L_{H_2} . When $\mathcal{A}_\infty$ requests an H_2 query: $\mathcal{B}$ checks if it is present in L_{H_2}. If not, $\mathcal{B}$ generates h_{2i} value, sets $h_{2i} = H_2(PID_i, K_{pub}, N_i, T_i)$, returns h_{2i} to $\mathcal{A}_\infty$.

H_3**Query**: $\mathcal{B}$ maintains a list L_{H_3} . When $\mathcal{A}_\infty$ requests an H_3 query: $\mathcal{B}$ checks if it is present in L_{H_3}. If not, $\mathcal{B}$ generates h_{3i} value, sets $h_{3i} = H_3(m_i, PID_i, PK_i, \bigtriangledown, t_i)$. $\mathcal{B}$ returns $h_{3i})$ to $\mathcal{A}_\infty$.

Partial private key query:if $PID_i \neq PID_i^*$, $\mathcal{B}$ first checks whether PID_i is in L_u.If it not, $\mathcal{B}$ obtains $ppk_i = n_i + kh_{2i}$ and returns it to $\mathcal{A}_\infty$. If $PID_i = PID_i^*$, $\mathcal{B}$ aborts the game.

Public Key Query:If $PID_i \neq PID_i^*$, $\mathcal{B}$ first checks whether PID_i is in L_u. If it not, $\mathcal{B}$ obtains $PK_i = (U_i, N_i)$ and returns to $\mathcal{A}_\infty$.

Public Key Replacement Query: $\mathcal{B}$ manages a list of L_{PK}. $\mathcal{B}$ replaces the public key with $PK_i^* = (U_i^*, N_i^*)$, $\mathcal{B}$ sets $N_i = N_i^*$, $U_i = U_i^*$, $SK_i = SK_i^*$, $B_i = \perp$, and inserts $(PID_i, n_i, N_i, SK_i, U_i)$ into L_{PK}.

Signature Query: When $\mathcal{B}$ receives a signature query from $\mathcal{A}_\infty$ for (PID_i, m_i): $\mathcal{B}$ first checks if PID_i is in L_{PK}. If it is, $\mathcal{B}$ selects $x, y, z \in Z_q^*$ and calculates $Y_i = x$, $R_i = (x - y - z)P$, $h_{3i} = H_3(m_i, PID_i, PK_i, \bigtriangledown, t_i) \leftarrow x^{-1}$, inserts $(m_i, PID_i, PK_i, \bigtriangledown, t_i, h_{3i})$ into L_{H_3}, and returns the signature $\delta_i = (R_i, Y_i)$ to $\mathcal{A}_\infty$.

Based on the above operations, the advantage of $\mathcal{B}$ addressing this paper successfully is $(1/q)(1 - q_H^2)(1 - q_H q_C/q)\varepsilon$. Therefore, the assumption is not valid, and we proved that the scheme can resist the forgery attack of the type 1 adversary $\mathcal{A}_\infty$ in ROM.

The proof process of **Theorem 2** is similar to **Theorem 1**, and finally we can get: the advantage of $\mathcal{B}$ addressing this paper successfully is $(1/q)(1 - q_H^2)(1 - q_H q_C/q)\varepsilon$. Therefore, the assumption is not valid, and we proved that the scheme can resist the forgery attack of the type 2 adversary $\mathcal{A}_\in$ in ROM.

B. Computational Overhead Analysis

The specific running time of each cryptographic operation in the experiment is shown in the table I.

TABLE I
EXECUTION TIME OF VARIOUS BASIC OPERATIONS

Notations	Description	Run time(ms)
T_p	Bilinear pairing	3.176
T_{bp-sm}	Bilinear pairing scalar multiplication	1.883
T_{ecc-sm}	Elliptic curves scalar multiplication	1.139
T_{hmtp}	Map to point hash function	4.694

The Fig.2 shows a comparison of the computational costs for the signature generation and signature verification stages among different schemes.

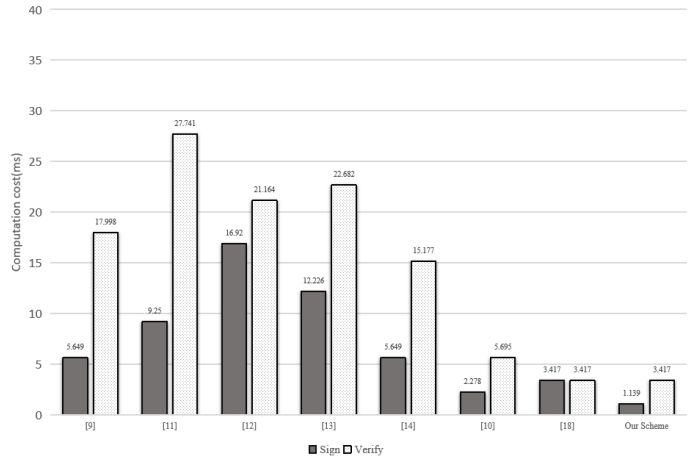

Fig. 2. Computational overhead of the personal signature generation and signature verification phases

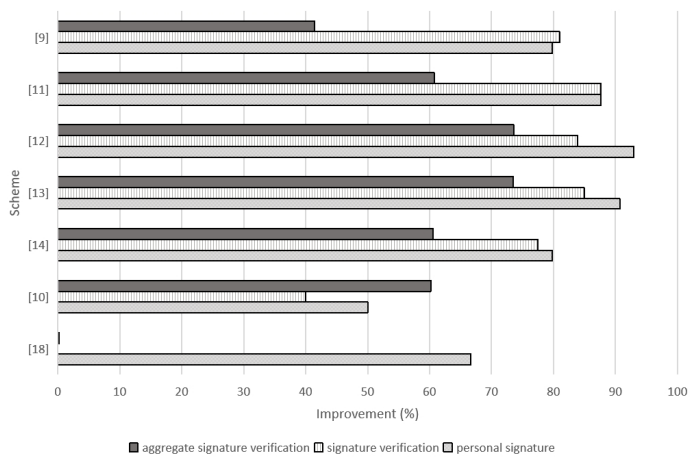

Fig. 3. Computational overhead of the personal signature generation and signature verification phases

The Fig.3 illustrates the percentage improvement in computational cost for three different phases of the scheme in this paper: signature phase, signature verification phase, and aggregated signature verification phase. It provides a visual representation of the improvement achieved in terms of computational efficiency for each phase. The schemes [9], [11]–[14] are all based on bilinear pairs and utilize scalar multiplication operations, bilinear pairing, and mapping to point hashing operations. However, previous research indicates that bilinear pairing can be time-consuming. Additionally, while [10], [18] and this scheme are based on ECC, the scheme presented in [10] is not more efficient than the scheme proposed in terms of computational cost. The scheme in [18] shares the same signature verification stage as our proposed scheme but is less efficient in the other two stages. Consequently, the proposed scheme has significant advantages in the analysis stage.

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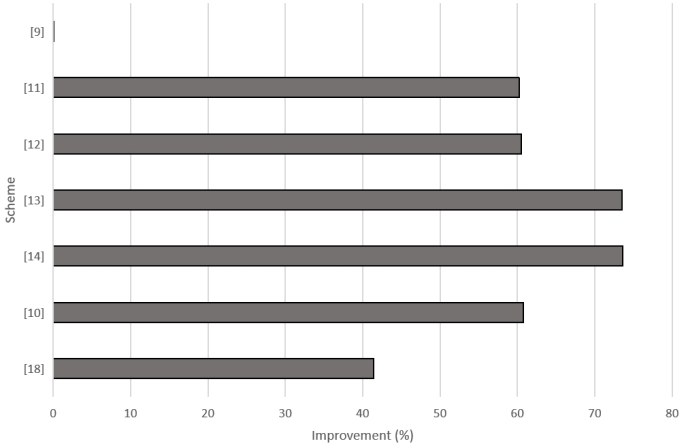

Fig. 4. Computational overhead of the personal signature generation and signature verification phases

C. Communication overhead analysis

The hardware environment we used for software simulation was an Intel i5-7200U processor with a clock frequency of 2.70GHz, 8GB of RAM, and the miracl library running on Windows 10 operating system. Compare references, the percentage increase in communication overhead of the scheme in this paper is shown in the Fig.4.

The comparison results demonstrate that the proposed scheme in this paper provides comprehensive security guarantees.

V. Conclusion

In this paper, we present a novel certificateless aggregate signature scheme with a tailored revocation mechanism for telemedicine applications. Our scheme aims to revolutionize security, privacy, and information management in telemedicine, offering practical implementation potential. The fusion of certificateless aggregate signature technology with cross-domain authentication ensures the security and integrity of patient medical data. The introduction of a revocation function addresses evolving patient privacy needs. Thorough security and performance analyses confirm the solution's safety, reliability, and efficiency, providing additional guarantees like traceability and revocability. Reduced computing and communication costs, as revealed in the performance analysis, underscore the scheme's practical viability. To extend our insights, future research could explore the scheme's applicability in diverse healthcare settings and consider enhancements such as integrating emerging technologies like blockchain. Furthermore, investigations into user acceptance and ethical considerations would contribute to ongoing advancements in secure telemedicine systems.

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